Pollution comes in many forms, ranging from toxins in the air to trash at the bottom of the sea. Any substances or forms of energy that spoil the quality of the atmosphere, oceans, water, or soil are pollutants. They may be chemicals or biological contaminants (including human waste), products (such as plastics), or noise, light, or heat. The effects of pollution on life of all kinds can be far-reaching, spreading thousands of miles beyond its original source. Pollution can spread through the food chain and be carried through air and water, affecting all life. Contaminants such as plastics can facilitate the invasion of nonindigenous species, as discovered by Australian marine biologist Emma Johnston. There is also a direct effect on human health: it is estimated that exposure to polluted air, water, and soil caused 9 million premature deaths—one in six of all deaths—in 2015.